

Assignment 7: Expressions

DO NOT WRITE ON THIS PAPER

Part A

Complete the bottom two rows of the table, then answer the questions below.

Expression: Number of 5's plus 19

Number of 5's	Factored Form	Expanded Form	Value
1	$5(1) + 19$	$5 + 19$	24
2	$5(2) + 19$	$5 + 5 + 19$	29
3	$5(3) + 19$	$5 + 5 + 5 + 19$	34
4			
5			

Based on the pattern you see in the table, use a variable **n** to write an expression for the sum of **n** 5's plus 19.

Part B

For each expression, use the given mathematical operation to write an equivalent expression.

- Distribute:** $7(3x+4)$
- Combine Like Terms:** $3x + 9 - 7x - 6 + 2x$
- Factor:** $6x + 15$

Part C

Consider the two expressions:

$$\frac{4(3x - 5x + 3)}{8} \quad \text{and} \quad \frac{-4x + 6}{4}$$

In the table below, evaluate both expressions using the given values of x. Write your answer as a fraction in simplest form.

X	Value of Expression 1	Value of Expression 2
$x = 0$		
$x = -1$		
$x = 3$		

Based on this table, are these two expressions likely equivalent?

Question 1

Big Question: What are we allowed to do with an expression?

Use the chart below to simplify the expression.

$$3(2x + 5 - 8x) + 5x$$

Write your work step by step and then explain each step in words. Your explanations should use the following vocabulary words: *combine like terms*, *distribute*

Note: You might not use all rows of the table.

Steps	Explanation
$3(2x + 5 - 8x) + 5x$	Given in the problem

Question 2 Work

Stephan is trying to determine how much it will cost to take his friends and himself to the movies.

Each ticket costs 8 dollars. Additionally, he wants to buy a bucket of popcorn to share with everyone, which costs 7 dollars.

Write an expression to determine the cost (in dollars) to bring his friends **and himself** to the movies.

Use **n** to represent the number of friends he brings.

Question 2 Answer

Write your answer
in your Class Notes

Question 3 Work

Evaluate the expression when $a=9$ and $b=8$.

$$7b + a + 3$$

Question 3 Answer

Write your answer
in your Class Notes